

Netflix Content Analytics Report

Project Overview

This project presents an end-to-end Netflix Content Analytics case study built using an ELT (Extract, Load, Transform) approach to answer key business questions related to directors, genres, countries, and content trends. The analysis identifies versatile directors, evaluates country-wise comedy movie production, determines top directors by year, analyze average movie duration across genres, and discovers directors who have worked across multiple genres. The project demonstrates practical data cleaning, transformation, and analytical SQL techniques to generate actionable business insights from real-world streaming data.

Dataset Summary

- Rows: 3,900

- Columns: 18

- Missing Data:

2637 values in Director column

825 values in Cast column

831 values in Country column

10 values in Date_added column

Data Extraction

We began with data extraction and initial inspections in Python

Data Extraction: Extracted data using Kaggle API

```
● √ import pandas as pd
    from kaggle.api.kaggle_api_extended import KaggleApi

    api = KaggleApi()
    api.authenticate()

    print("Kaggle API connected successfully")

Kaggle API connected successfully
```

Data Loading

Imported data in Pandas

```
api.dataset_download_files(  
    'shivamb/netflix-shows',  
    path='./data',  
    unzip=True  
)  
  
print("Successfully imported")
```

Dataset URL: <https://www.kaggle.com/datasets/shivamb/netflix-shows>
Successfully imported

```
df = pd.read_csv('data/netflix_titles.csv')  
  
print('Datasets loaded successfully')
```

Datasets loaded successfully

Initial Exploration: Used `df.shape()` to check structure and Checked null values using `df.isnull().sum()` .

```
df.info()
```

✓ 0.1s

```
<class 'pandas.DataFrame'>  
RangeIndex: 8807 entries, 0 to 8806  
Data columns (total 12 columns):  
#   Column          Non-Null Count  Dtype  
---  ---  
0   show_id         8807 non-null   str  
1   type            8807 non-null   str  
2   title           8807 non-null   str  
3   director        6173 non-null   str  
4   cast            7982 non-null   str  
5   country         7976 non-null   str  
6   date_added      8797 non-null   str  
7   release_year    8807 non-null   int64  
8   rating          8803 non-null   str  
9   duration        8804 non-null   str  
10  listed_in       8807 non-null   str  
11  description     8807 non-null   str  
dtypes: int64(1), str(11)  
memory usage: 825.8 KB
```

```
df.isnull().sum()
```

show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0
dtype: int64	

Data Transformation

Performed integration of python and sql using `sqlalchemy pymysql`

```
import pymysql
from sqlalchemy import create_engine

engine = create_engine(
    "mysql+pymysql://root:spatil123@localhost:3306/netflix"
)

print("Connected Successfully")
```

Connected Successfully

Now created `netflix` schema in sql

Created raw tables for importing data in sql database

```
-- Creating raw table for importing data in sql database
CREATE TABLE netflix_raw (
    show_id varchar(20),
    type varchar(20),
    title nvarchar(150),
    director nvarchar(250),
    cast nvarchar(800),
    country nvarchar(150),
    date_added varchar(20),
    release_year int,
    rating nvarchar(20),
    duration nvarchar(20),
    listed_in nvarchar(100),
    description nvarchar(300)
);
```

Append Data in SQL Table

```
df.to_sql(
    name='netflix_raw',
    con=engine,
    if_exists='append',
    index=False
)

print("Data Loaded Successfully")
```

Data Loaded Successfully

Created a query to finding duplicates values

```
--
23 • Select * from netflix_raw where concat(title, type) in (
24     Select concat(title, type)
25     from netflix_raw group by title, type
26     having count(*) > 1) order by concat(title, type); #This query finds duplicate values
27
```

Result Grid

show_id	type	title	director	cast	co
s304	Movie	Esperando la carroza	Alejandro Doria	Luis Brandoni, China Zorrilla, Antonio Gasalla, Julio De Grazia, Betiana Blum, Monica Villa, Juan Manuel Tenuta, Andrea Tenuta, Cecil...	Arq
s6706	Movie	Esperando La Carroza	Alejandro Doria	Luis Brandoni, China Zorrilla, Antonio Gasalla, Julio De Grazia, Betiana Blum, Monica Villa, Juan Manuel Tenuta, Andrea Tenuta, Cecil...	Arq
s160	Movie	Love in a Puff	Pang Ho-cheung	Miriam Chin Wah Yeung, Shawn Yue, Singh Hartihan Bitto, Isabel Chan, Cheung Tat-ming, Matt Chow, Chui Tien-you, Queenie Chu,...	Hor
s7346	Movie	Love In A Puff	Pang Ho-cheung	Miriam Chin Wah Yeung, Shawn Yue, Singh Hartihan Bitto, Yat Ning Chan, Tak-Ming Cheung, Matt Chow, Chui Tien-you, Queenie C...	Hor
s1271	TV Show	Sin senos sí hay paraíso	NULL	Catherine Sciocheque, Fabián Ríos, Carolina Gaitán, Juan Pablo Urrego, Majida Issa, Johanna Fadul, César Mora, Juan Pablo Llano, ...	Uni
s8023	TV Show	Sin Senos sí Hay Paraíso	NULL	Majida Issa, Fabián Ríos, Catherine Sciocheque, Carolina Gaitán, Juan Pablo Urrego, Francisco Bolívar, Johanna Fadul, Jennifer Are...	Uni
s5026	Movie	Veronica	Paco Plaza	Sandra Escacena, Claudia Placer, Bruna González, Iván Chavero, Ana Torrent, Consuelo Trujillo	Spt
s5250	Movie	Verónica	Carlos Algara, Alejandro Martinez-Beltran	Arcelia Ramirez, Olga Segura, Sofia Garza	Me

Created a new tables for Directors, Country, Listed in & cast for performing normalization

```
-- Tables for directors
create table netflix_directors (
    SELECT
        n.show_id,
        TRIM(jt.director) AS director
    FROM netflix_raw n
    CROSS JOIN
    JSON_TABLE(
        CONCAT( '[', REPLACE(REPLACE(n.director, '"', '\\\"'), ',', ', ', '"', ''), ']' ),
        '$[*]' COLUMNS (director VARCHAR(255) PATH '$')
    ) AS jt
    WHERE n.director IS NOT NULL
    AND n.director <> '' );
```

Created final tables for Data analysis

```
-- Creating final table for analysis
Create table Netflix_data as
with cte as (
    Select *, rank() over(partition by title, type order by show_id) as rnk from netflix_raw
)
select show_id, type, title, str_to_date(date_added, '%M %d, %Y') date_added, release_year, rating,
(case when duration is null then rating else duration end) as duration, description
from cte where rnk =1; #This is the final table
```

Data Analysis in SQL

1. Identify directors who have successfully produced both Movies and TV Shows, helping Netflix recognize versatile content creators.

Result Grid			
		Filter Rows:	Export
	director	Movie_count	TV_show_count
▶	Abhishek Chaubey	4	1
	Alastair Fothergill	1	3
	Alban Teurlai	1	1
	Alessandro Angulo	1	1
	Andrew Tan	1	1
	Anurag Kashyap	8	1
	B. V. Nandini Reddy	3	1
	BB Sasore	2	1
	Bejoy Nambiar	2	1
	Billy Corben	1	1
	Brad Anderson	3	1
	Bunmi Ajakaiye	1	1
	Cosima Spender	2	1
	Dan Forrer	1	1
	Daniel Kontur	1	1
	David Ayer	3	1
	Elaine McMillion Sh...	2	1
	Eli Roth	2	1
	Estela Renner	1	1
	Faris Godus	1	1
	Gary Howsam	1	1
	Hirozuki Seshita	3	1

2. Determine which country has produced the highest number of Comedy Movies, enabling regional content performance analysis.

Result Grid		
		Filter Rows:
	Country	no_of_comedy_movies
▶	United States	896
	India	329
	United Kingdom	112
	Canada	96
	Turkey	58
	Egypt	57
	France	56
	Spain	48
	Germany	47
	Mexico	42
	Nigeria	41
	Hong Kong	32
	China	31
	Brazil	29
	Philippines	28
	Argentina	21
	Belgium	20
	South Korea	19
	Australia	18
	South Africa	18
	Italy	16
	Indonesia	15

3. Identify the top-performing movie director each year (based on the date content was added to Netflix) to understand trends in content acquisition.

	date_added	director	no_of_movies	rnk
▶	2021	Rajiv Chilaka	17	1
	2021	Suhas Kadav	15	2
	2018	Raúl Campos	12	1
	2018	Jan Suter	12	2
	2020	Youssef Chahine	11	1
	2017	Umesh Mehra	7	1
	2017	Jay Chapman	7	2
	2018	Johnnie To	7	3
	2019	Martin Scorsese	7	1
	2019	Cathy Garcia-Molina	7	2
	2021	Hidenori Inoue	7	3
	2017	Marcus Raboy	6	3
	2017	S.S. Rajamouli	6	4
	2017	Tilak Shetty	6	5
	2018	Matt Askem	6	4
	2018	Wong Jing	6	5
	2018	Riri Riza	6	6
	2019	Wenn V. Deramas	6	3
	2019	Kunle Afolayan	6	4
	2019	Steven Spielberg	6	5
	2020	Cathy Garcia-Molina	6	2
	2020	Joey So	6	3

4. Analyze the average duration of movies across different genres to better understand audience content preferences and genre characteristics.

	listed_in	Average_duration
▶	Classic Movies	118.6466
	Action & Adventure	113.5157
	Dramas	113.0685
	Romantic Movies	110.5731
	International Movies	110.3689
	Thrillers	107.1664
	Music & Musicals	106.1253
	Sci-Fi & Fantasy	106.0165
	Faith & Spirituality	105.5846
	Cult Movies	104.6571
	Comedies	103.8822
	Independent Movies	101.0781
	Horror Movies	98.4454
	Sports Movies	94.6986
	LGBTQ Movies	94.2941
	Anime Features	92.9437
	Documentaries	81.6306
	Children & Family M...	79.9360
	Stand-Up Comedy	67.3265
	Movies	47.0175

5. Find directors who have created both Comedy and Horror movies, highlighting creators with diverse filmmaking expertise.

	director	no_of_comedies	no_of_horror
►	Adam Egypt Mortimer	1	1
	Ahmed Zein	2	2
	Álex de la Iglesia	2	1
	Amar Kaushik	1	1
	Anggy Umbara	1	1
	Anthony Scott Burns	1	2
	Anurag Kashyap	1	1
	B.N. Shajeer Sha	1	1
	Banjong Pisanthanakun	1	3
	Brad Peyton	1	1
	Brent Maddock	1	1
	Dennis Widmyer	1	1
	Dibakar Banerjee	3	1
	Don Michael Paul	3	3
	Eli Craig	2	2
	Gary Shore	1	1
	Gore Verbinski	1	1
	Hardik Mehta	1	1
	Jason Krawczyk	1	1
	Jeff Baena	2	1
	Jeremy Saulnier	1	2
	Joe Dante	1	1

Business Recommendations

1. Collaborate with directors who create both Movies and TV Shows to diversify content offerings.
2. Increase investment in countries with the highest Comedy Movie production to strengthen regional content.
3. Build long term partnerships with top performing directors identified each year.
4. Use genre wise average movie duration insights to optimize future content production.
5. Encourage collaborations with directors experienced in both Comedy and Horror to expand multi-genre content.